TABLE S6a

PAIRED REFERENCE AGREEMENT

Clustering	Model	Means (%)	Means + fractions (%)	Paired change (pp)
Leiden	GPT-6 Astra	52.86	58.12	+5.26
FlowSOM	GPT-6 Astra	36.66	36.05	-0.61
SpatialSort	GPT-6 Astra	36.13	32.38	-3.75
PIXIE	GPT-6 Astra	41.32	35.64	-5.68
Leiden	Claude Opus 5.5	46.78	50.27	+3.49
FlowSOM	Claude Opus 5.5	37.48	36.82	-0.66
SpatialSort	Claude Opus 5.5	33.11	36.17	+3.06
PIXIE	Claude Opus 5.5	36.04	42.54	+6.50
Leiden	Gemini 4 Argon	46.16	47.51	+1.35
FlowSOM	Gemini 4 Argon	39.57	41.84	+2.27
SpatialSort	Gemini 4 Argon	33.31	33.80	+0.49
PIXIE	Gemini 4 Argon	38.70	36.40	-2.30
Leiden	DeepSeek V4.1 Flash 1017	50.17	39.92	-10.25
FlowSOM	DeepSeek V4.1 Flash 1017	27.62	27.87	+0.24
SpatialSort	DeepSeek V4.1 Flash 1017	20.25	17.42	-2.83
PIXIE	DeepSeek V4.1 Flash 1017	33.42	27.35	-6.07

Completed full-arm bank; equal weighting of two test regions; fixed 28-label vocabulary including Noise. One accepted annotation per cluster and condition; no repeated-call SD.